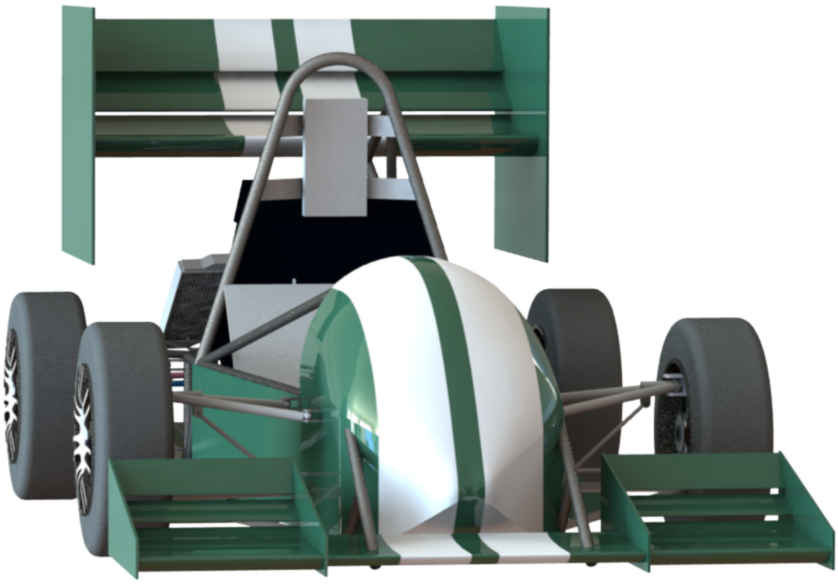
Imperial College London – Department of Aeronautics

Front Wing Design

Group Design Project : Imperial Racing Green 2017



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Abstract

The objective of this group design project was to design a four-wheel drive, electric racing car (EV4) to compete in the 2017 Formula Student event, whilst achieving as highly in the competition as possible. It became apparent when considering the budgetary constraints of the task that to fulfil the four-wheel drive criterion was impossible. As such, a competitive rear-wheel drive concept was designed instead to meet the remaining requirements. This report series details the initial design phase of a 2 wheel-drive, open-cockpit Formula Student car, conducted by the Department of Aeronautics at Imperial College London for the purpose of presenting the Imperial Racing Green team with a design they can implement over a number of iterations. The final design has a mass of 252 kg (excluding driver) and will cost £16,463 to implement, and is estimated to score within the top 10 of the FSUK competition. All the design aspects of the vehicle have complied with the regulations and the technical inspection forms alongside other documentation have been considered.

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Executive Summary

Introduction

This project is an early stage design proposal for a competitive open-wheel, open-cockpit, Formula Student race-car for implementation by the Imperial Racing Green team. Due the team's limited annual budget, the design is recommended for implementation over a number of iterations, and can be done so for an estimated £16,463.
The Formula Student UK (FSUK) competition takes place in July on the Silverstone Circuit, with 107 in attendance in 2016. It is organised by the Institute of Mechanical Engineers and consists of 4 static and 5 dynamic events. The Imperial Racing Green team's current car has not yet passed the required Technical Inspection to race in any previous year, and so the design outlined below is based on principles of simple, robust design, and strict adherence to the FSUK regulations. It is estimated to achieve placement within the top 10 at the 2016 competition's dynamic events, at a fraction of the cost of the highest performing teams.

For detailed information regarding the Formula Student Competition, see report written by Puja Dodhia. For detailed information regarding the Project Management, see report written by Kieran Downie.

Performance Estimation

The lap simulator is an important tool used to aid design decisions. The simulator models the performance of the car in each dynamic event of the competition and estimates the performance of the car according to the points metrics defined by the FSAE. The simulator assumes the driver maximises the grip of the tyres in every traction and braking zone, and this 'perfect driver' assumption is the main source error in short duration events such as acceleration and skid pad, which have the largest error, but predicts the performance of previous Formula Student entries extremely accurately (to within 1%) for the endurance and efficiency events.
Once completed, the two-wheel drive (2WD) Lap Simulator had to be validated against real-world vehicles to ensure that it was a reliable tool during the design process. Three teams were found that had similar configurations to the Lap Simulator model: Universitat Politecnica de Catalunya 2016, RWTH Aachen University 2015, and TU Darmstadt 2015. Comparing the actual scores and simulated scores, the 2WD Lap Simulator is able to predict the overall scores of the three sample cars with an average percentage error of 6.4%.

For detailed information regarding the Performance Estimation, see report(s) written by Gao Neo & Alexander Roberts.

Global Vehicle Layout

To compete in the FSUK competition, EV4 had to comply with the defined Rules & Regulations. Additionally, maximising the car's performance was a driving factor for the component packaging. The shortest wheelbase imposed by regulations (1525 mm) was selected along with a trackwidth of 1300 mm to improve the car's handling and reduce weight. The car's performance was shown to be highly sensitive to mass (0.8 points won per kg lost) so mass savings were a priority. A final weight of 252 ± 10kg was achieved which is level with the top FSUK teams with similar configurations. Furthermore, heavy elements such as the driver, battery and motor controllers were placed on the bottom of the chassis to obtain a low CG position of 314 mm.
The initial configuration consisted of 4 in-wheel motors, with a tubular space frame chassis and a full aerodynamic package. To increase the car's performance, reduce weight, and stay in the team's budget, a RWD configuration was adopted. To compensate for the resulting rearward shift in mass, it was decided to move the chassis forward by 200 mm relative to the wheels, achieving a final static weight distribution of 44/56%. The final longitudinal position of the CG is 1646 mm from the foremost point on the car and aligned along the vehicle's centerline.

For detailed information regarding the Global Vehicle Layout & Mass Estimation, see report(s) written by Cesar Maklary & Wenxi Chen.

Chassis Design

The chassis of the vehicle was designed in order to achieve efficient structural integrity, whilst minimising mass. Based on these criteria, the configuration of the chassis was chosen to be steel spaceframe, which is also low cost and amenable to changes in an iterative design. The chassis was tested using FE analysis to determine whether the structure would yield under the loading specified by the regulations and the loading experienced during the dynamic events. The results show the chassis is structurally capable and can bear all the applied loads without yielding or buckling. The total cost of the chassis is £251.86. The final mass of the chassis is 31 kg, which is a 40% decrease from the first iteration.

For detailed information regarding the Chassis Design, see report(s) written by Salomon Livingston & Zhehong Zhang.

Suspension Design

The suspension was kept as an unequal length double wishbone with pullrod configuration. Working with both the chassis and upright design teams to create structurally sound mounting points for the suspension, anti-squat and anti-dive features were included. With the suspension geometry finalised, wheel travel, camber gains and spring travel were all simulated in a kinematic model with MATLAB. Therewith, it was possible to choose suitable spring-rates. A 4-stage monoshock damper was used to adjust the damping ratios. Lastly, an anti-roll bar was included on the front axle, which meant the roll stiffness of the car could be adjusted to suit the drivers preferences. An anti-roll bar could not be fitted on the rear axle due to space constraints with the battery. Using the aforementioned MATLAB kinematic model, the suspension was implemented in the lap simulator. The point-score gain was minimal, at $+0.02$ points. It was believed this was due to the simulator's "perfect driver" assumption, given that the benefits of a good suspension design come in driveability and thus, these benefits cannot be modelled. Therefore, greater gains would be expected on track with a human driver. 4130 steel was chosen, giving a total suspension mass of 6.85 kg. Carbon fibre was considered, however, the weight saving of approximately 2 kg (which converts to roughly 1.6 points) was not worth the additional costs and manufacturing complexity involved.

For detailed information regarding the Suspension Design, see report(s) written by Sean Cartmale.

Powertrain Design

The powertrain design aimed to package the drivetrain and brake components as efficiently as possible within the rear wheels. Drivetrain mass reduction was also a key factor with unsprung weight influencing the car's handling. By mounting the gearbox in the centre of the brake disk along with careful component selections, the drivetrain is neatly packaged within the wheel. A 1:7 gear ratio was chosen, maximising the dynamic event points scored. Based on packaging, mass and operating conditions constraints the Apex Dynamics AD090 gearbox was selected. Both the front and rear upright designs accommodate the suspension and ackerman steering geometry, and are designed to withstand the worst case loading scenarios.
A permanent magnet DC motor EMRAX 188 was chosen for its high maximum torque of 100Nm and no-load speed of 7900 RPM. A high maximum torque ensures greater maximum acceleration while a high no-load speed gives a higher maximum speed of the vehicle. Maximum acceleration and speed of the car are both important in scoring as many points as possible in dynamic events. The high efficiency of the EMRAX 188 motor is also favourable, reducing heat loss and hence cooling required. Moreover, regenerative braking could be incorporated in future improvements. Due to budget and power constraints, it is suggested that EV4 adopts a rear-wheel-drive configuration. The required torque to provide sufficient acceleration is satisfied by the chosen EMRAX 188 motors in RWD. The lap simulator has also been modified to account for this change.

For detailed information regarding the Powertrain Design, see report(s) written by Elis Jackson & Pak Ho Wan.

Battery Selection

The battery is an essential part of the powertrain system to satisfy the power requirements for efficient propulsion of the vehicle. By working closely with the lap simulator, propulsion performance and systems integration teams, the necessary power and energy requirements were calculated to compete in the endurance race from which the battery design was based. A total of 126 EiG C020 Li-Ion pouch cells arranged in a 21-series-6-parallel configuration were selected to power 2x EMRAX 188 motors and secondary components. Sub-components were defined as part of the reliability study and effects of battery heating were considered to ensure prolonged battery life. The battery casing has been designed in accordance with FSAE regulations with regards to minimising the volume and achieving the optimal weight distribution. A structural analysis was conducted to ensure compliance with the regulations. The overall weight of the battery system was found to be 65.3 kg which comprises of the battery cells, container & sub-components weighing 65.3 kg in total.

For detailed information regarding the Battery Selection & Integration, see report(s) written by Nikita Vorobiov.

Aerodynamics

An aerodynamic package has developed, consisting of front and rear wings, diffuser and bodywork, utilising Computational Fluid Dynamics in Star CCM. Following the results of a sensitivity study, it was found that the final score increased significantly with SCL (dimensional downforce coefficient) and decreased with SCD (dimensional drag coefficient). Thus the design was aimed at maximising the downforce-to-drag ratio, L/D, of the full car. Simulations run on the car with the final aerodynamic package estimate the total downforce to be 557 N and a total drag of 214 N, averaged over a lap. The aerodynamic package is estimated to weigh a total of 25.3 kg. The lap simulator showed that using an aerodynamic package produced a score increase of 36.4 points. The aerodynamic balance was found to be 40% - 60% rear-biased, which benefits the car's RWD configuration.

Front and Rear Wing

The front wing was designed to maximise L/D. The final design consists of 3-elements: 1 mainplane and 2 flaps. When viewed from the front, the flapped sections cover the front wheels and extend inwards towards the body, leaving gaps of 65 mm from the body to allow airflow through for improved diffuser performance and to aid in cooling. Tapered endplates were chosen to save weight as rectangular endplates did not show increased performance. The front wing will be attached to the chassis via two pylons, which will have a member braced between them to reduce the sideways deflections. Overall, the front wing produces a downforce of 186 N at a drag penalty of 36 N and weighs 7.5 kg. The rear wing was also designed to maximise L/D whilst maintaining enough downforce to be beneficial to the overall performance. The final design was triple-element with a large surface area to maximise downforce, even at relatively low velocities. By elevating and shifting the wing rearward, turbulence from upstream geometries was minimised. Rectangular endplates were most aerodynamically efficient and were extended to reduce vortex shedding and stabilise the turbulent wake from the tires. The final design was tested using CFD, and was found to produce 219 N of downforce for a 67 N drag penalty, weighing 8kg. The rearward aerodynamic bias was deemed beneficial due to the need for loaded rear wheels for maximum traction. A triangulated steel frame welded to the chassis supports the wing and is attached via a metal plate.

For detailed information regarding the Front Wing Design, see reports written by Tze Tiong & Abhishek Aiyer.
For detailed information regarding the Rear Wing Design, see reports written by Matt Gillie & Sean Cartmale.

Diffuser and Undertray

The undertray acts to accelerate the flow under the vehicle, reducing the pressure and increasing the downforce. The diffuser is then used to expand the air flow back to ambient pressure. The underbody experiences the lowest drag and has the highest downforce-to-drag ratio amongst all the aerodynamic components on EV4. The geometry of the underbody was restricted by regulations and the dimensions of other components (eg. chassis, wheel track, wheel base and jacking bar). By sweeping the design parameters, a final design was reached, producing 152N of downforce and 6.3N of drag, weighing 9.8kg. Considering the cost and weight penalty, E-grade glass fibre with a thickness of 2mm was chosen for the underbody.

For detailed information regarding the Diffuser and Undertray Design, see report(s) written by Weihao Lu.

Bodywork Design

The objective of the bodywork was to cover the internal structure of the vehicle and provide an aerodynamic profile. To design the latter, a larger focus was placed on reducing drag, rather than improving downforce. In order to achieve this, a MATLAB code was created which generated a bodywork curvature which minimised frontal area and ensured a smooth curvature throughout. The final design weighs 10kg, produces 22 N of lift, and 31 N of drag. Polystyrene (PS) was chosen as the material for the bodywork as it has the lowest cost and density merit function, and it had a sufficiently high Young's Modulus to withstand the forces acting on it.

For detailed information regarding the Bodywork Design, see report(s) written by Bryan Ong.

System Layout & Design

Steering System

The objective of the design was to enable the steering system to adequately control the car whilst also fitting within the chassis. A front wheel drive steering system was chosen due to its reduced complexity and effectiveness in steering at high speed. For the steering rack, a rack and pinion system was chosen as it is light and has high steering accuracy. For the steering geometry design, an Ackermann steering model was used. The Ackermann model represents the ideal steering angle with no tyre scrubbing, where all 4 wheels corner about the same point. An optimal steering geometry was obtained using an iterative MATLAB code.

For detailed information regarding the Steering System, see report(s) written by Bryan Ong.

Braking System

This was designed to achieve a maximum deceleration of 1.5g. An iterative method was used to select and size the various subcomponents of the system. The final system consists of an in-house designed pedal box with a pedal ratio of 3.3:1, 2 AP Racing CP2623 master cylinders, 4 AP Racing CP3696-6E0 brake calipers, and 4 in-house designed brake discs. A common manufacturer for most of the components allows easy integration. The brake system is designed to lock up all 4 wheels at the same time with an application force of 270 N. The distribution of braking force is approximately 75:25 front to rear axle. A finite element analysis was conducted on the design of the brake disc based on a loading condition of 2000 N on the brake pedal and at a temperature of 400°C. Moreover, a thermal analysis was used to determine the maximum temperature and the temperature distribution in the brake disc after one lap of the endurance event.

For detailed information regarding the Braking System, see report(s) written by Brandon Wong.

Cooling System

The endurance race was used to identify components in need of a cooling system. Components identified were the Electric Motors, Motor Controllers, Brake Discs and Battery packs. From thermal analysis of conduction, convection and radiation it was found that the brake discs and battery pack were cooled sufficiently without liquid cooling. The battery pack will not reach critical temperatures while the brake discs are cooled by the airflow. The electric motors and motor controllers will require a liquid cooling system comprised of 2 radiators, a water pump and catch can all used on the current Imperial Racing Green vehicle. The cooling system must have a power of 2050 Watts. To achieve this the radiators require an average velocity of air of approximately 10 m/s. CFD results for the entire vehicle was used to place the radiators where the airflow is sufficient.

For detailed information regarding the Cooling System, see report(s) written by James Marchant.

Torque Vectoring

The working principle behind a torque vectoring system is in applying different braking or tractive forces at each wheel to generate a torque about the vehicle, hence controlling the vehicle's yaw rate. This could be implemented using sensor-acquired inputs of steering angle and vehicle speed fed to a master controller which uses a torque vectoring algorithm to control the rotational speed of each wheel. The Munusamy & Weigl algorithm was implemented into the 2WD Lap Simulator. The torque vectoring was shown to give a significant benefit: reducing lap time be 0.4 seconds per lap, reducing energy consumption from 13.0kWh to 12.6kWh for the endurance event, with an increase of 10 points on the overall point-score for a cost of just £134.40.

For detailed information regarding the Torque Vectoring, see report(s) written by Brandon Wong & Gao Neo.

Driver Environment & Interfaces

Driver Ergonomics

The seat, head rest and cockpit designs mainly focused on maximising driver ergonomics, leaving sufficient space for packaging and compliance with the regulations. The placement of pedal box is 1000 mm from the bottom of the seat with 65 mm adjustment. Furthermore, based on the study of EV2 and driver ergonomics, the steering wheel is placed 500 mm from the bottom of chassis and 650 mm from the driver's shoulder to provide a comfortable reach position. The seat back angle was 45^o for driver comfort inside the frame which also satisfies the regulations.

For detailed information regarding the Driver Ergonomics, see report(s) written by Cesar Maklary & Wenxi Chen.

Electronics & Control

The electrical system acts as a relay between the driver and the car. The sensors monitor the throttle pedal angle, steering angle and wheel speed. This information is passed onto the Arduino Due micro-controller which processes the signal before sending it to the motor controllers. The motor controllers regulate the amount of current and voltage drawn from the battery to power the motor, based on the signals from the micro-controller. 2 Emdrive 500 motor controllers were selected to output a continuous voltage of 90V and 400A of current to power 2x EMRAX 188 motors. Communication between the microcontroller and the motor controllers are through a CAN bus. The electrical system has a kill switch that allows the driver to switch off the electrical supply in an emergency.

For detailed information regarding the Sensors, see report(s) written by Brandon Wong.

Firewall Regulatory Requirements

The firewall must separate the driver from all high voltage tractive systems. It has to be rigid, non-permeable, fire-resistant, electrically insulating, and securely sealed to the chassis to prevent passage of fluids. To utilise space efficiently, the battery was placed under the seat. To pass the cockpit test, it had to pass below the upper side impact bar. It also had to protect up to 100 mm of the driver's head from the their neck. Therefore, the design had to follow the shape of the head restraint. The FR4 Fibreglass Epoxy chosen for the insulating layer was tested for its suitability, including a penetration test and flammability check before being implemented.

For detailed information regarding the Firewall Design, see report(s) written by Puja Dodhia.

Vehicle Stability

A bicycle model was used to carry out a vehicle dynamic analysis. The analysis started with a directional steady state analysis, followed by a transient analysis. For the latter, the previous group's mathematical model was used with Simulink state space to model the response. The vehicle was vertically stable and this was validated from lap sim accelerations which did not surpass the allowable acceleration and decelerations of $-29m/s^2$ and $18m/s^2$. This means that all 4 wheels will remain on the ground during the race. With the car displaying oversteer, it becomes unstable during cornering. The directional analysis showed that the car would be controllable to considerable margins, even with just a 'good driver' model. This was an open loop analysis and hence did not incorporate the vehicle/driver interaction. Although the vehicle was unstable, analysis showed it to be controllable because it never goes beyond the critical speed (57.8 m/s). This configuration yields a better performance.

For detailed information regarding the Stability Analysis, see report(s) written by Kutlo Popo.

Final Vehicle Specification & FSAE Competition Prediction

Table 1: Final Vehicle Specification

Component	Unit	Value/ Description
Wheelbase	mm	1525
Wheel Track	mm	1300
Weight	kg	252 ± 10
Downforce-to-Drag Ratio	-	2.59
Predicted Result (FSUK 2016)	-	2nd - 8th (of 107)
Total Cost of Implementation	£	16,463

For detailed information regarding the Final Design 3D Modelling, see report(s) written by Yang Ke & Zhehong Zhang.

Front Wing Design

1 Introduction

Success in the Formula SAE competition has been a major goal of the Imperial Racing Green ever since its inception. This year, the EV4 has been developed to achieve that. This paper details the design of the front wing, a major component of the complete aerodynamic package which consists of the front and rear wings, the diffuser and bodywork. The need for an aerodynamic package has been the subject of constant debate; many teams in the competition have chosen not to include one yet most of the top-performing teams have opted to. Theoretically, an aerodynamic package provides great performance improvements at a slight weight penalty: the increased downforce creates greater traction between the wheels and the ground, allowing for greater cornering speeds and hence reduced lap times [1]. In particular, the front wing serves mainly to balance the downforce generated by the rearward aerodynamic components as well as to divert air over the front wheels, reducing drag [2]. However, a quantitative method for justifying the use of an aerodynamic package was necessary in the form of a lap simulator code [3]; the scores from the lap simulator were used to direct the design process and specify a design goal, as detailed in later sections.

2 Regulations

Any car designed to compete in the Formula SAE competition has to satisfy certain regulations as found in [4]. The regulations that affect the design of the front wing are summarised as follows:

- (T9.2) The front wing must not be within 75 mm of the front wheels, must not exceed 700 mm in front of the front wheels and must not be wider than the outside of the front wheels. The portion of the front wing directly in front of the wheels must be within a height of 250 mm from the ground.
- (T9.5) All forward-facing horizontal edges must have a minimum radius of 5 mm while all vertical edges must have a minimum radius of 3 mm.
- (T9.7) A force of 200 N applied at any point must not result in a deflection of more than 25 mm.

3 Front Wing Design

To identify the design parameter that would optimise the entire aerodynamic package, an initial sensitivity study was carried out using the lap simulator code [5] to determine the major factors influencing the score achieved in the dynamic events. From the initial study, it was found that an average increase of 16.7 points would be achieved per SC_L with an average decrease of 23.0 points per SC_D (see Figure 1(a)), where S denotes the frontal area of the car and C_L and C_D denote the downforce and drag coefficients respectively – these dimensional coefficients were used to account for the large variations in drag that would result from large variations in frontal area, which would have gone unnoticed if non-dimensional coefficients were used instead. The study also revealed a linear decrease of 0.8 points per kilogram of mass added (as per regulations) and that the score was relatively insensitive to the Center of Pressure position, as shown in Figure 1(b).

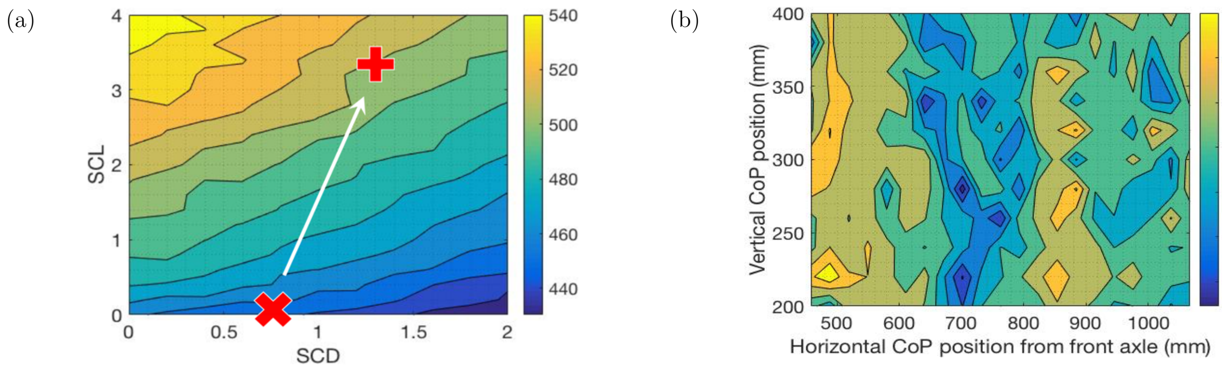

Figure 1: (a) Sensitivity of scores to SC_L and SC_D. The '×' shows the score estimate achieved using SC_L and SC_D values measured in a CFD simulation of the full car without an aerodynamic package. The '+' shows the score achieved when simulated with the final aerodynamic package of the EV4. (b) Sensitivity of scores to the Centre of Pressure (CoP) position. It can be seen that a large variation in the CoP position results in a difference of only a few points.

Based on the above results, the design of the front wing was aimed at maximising the downforce to drag ratio, L/D. This section details the steps taken to reach its final design.

3.1 Aerofoil Selection

Based on the lap simulator, the average speed of the car was calculated to be 16.5 m/s. This translates to a Reynolds number of approximately $5{\times}10^5$ for the mainplane and $1.5{\times}10^5$ for the flaps. Aerofoils with a minimum thickness of 10% (for structural reasons) were ranked by L/D in an aerofoil database [6] at the respective Reynolds numbers; four aerofoils each were then chosen for the mainplane and flaps. By importing the coordinates of the aerofoils into XFLR5, these were compared as a function of angle of attack as shown in Figures 2 and 3.

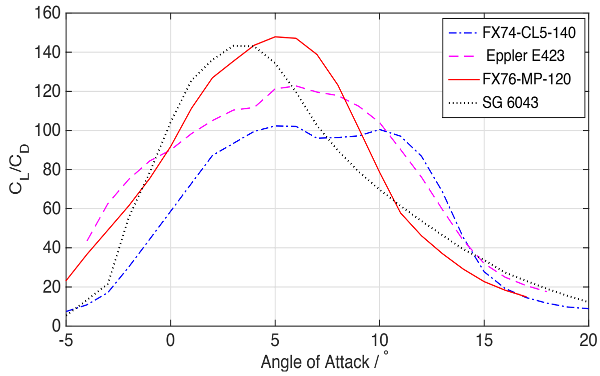

Figure 2: L/D variation with angle of attack plotted for the four candidate aerofoils for the mainplane.

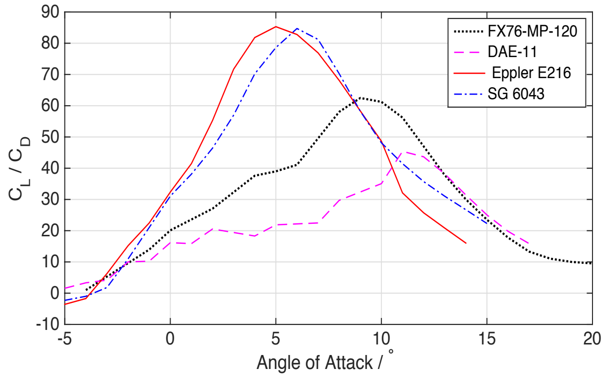

Figure 3: L/D variation with angle of attack plotted for the four candidate aerofoils for the flaps.

From the above figures, the FX76-MP-120 was chosen for the mainplane and the Eppler E216 was chosen for the flaps. Following the selection, the aerofoils were run again in XFLR5 at Reynolds numbers corresponding to the minimum (10 m/s) and maximum (20 m/s) speeds to ensure acceptable L/D performance at these speeds. Finally, the C_L vs angle of attack plots of the aerofoils were examined to check that the post-stall lift would decrease gradually; a sharp decrease in lift would pose a safety risk to the driver, resulting in a sharp loss of downforce and hence traction during cornering.

Based on Figure 2, the mainplane setting angle of 5.25° was also decided.

3.2 CFD Setup

Wind tunnel testing and computational fluid dynamics (CFD), in addition to track testing, constitute most of the aerodynamic testing of a Formula One car [7]. This allows for cross-validation between both computational and experimental approaches. However, due to the lack of resources and time in the design of the EV4, wind tunnel testing was not performed; CFD simulations using STAR-CCM+ were used as the main tool in making design decisions. Validation of the CFD solver using experimental aerofoil data was thus required and will be discussed below.

Due to limited time available, CFD simulations in 2-dimensions (2D) were preferred over simulations in 3-dimensions (3D) for design decisions that could be made using a cross-section of the front wing, such as setting angles and chord ratios; this allowed for a wider range of tests to be carried out. Components of the front wing that work in 3D, such as end plates, required simulations to be run on 3D models of the front wing.

3.2.1 Domain setup

As mentioned above, both 2D and 3D simulations were run on the front wing. In all simulations, the front wheels were included to account for the reduction in front wing performance due to interactions with the front wheel. This can be seen in Figure 4. Furthermore, while measurements of downforce were made on the front wing alone, all drag measurements included both the front wing and wheels. This was especially important in the comparison between 2- and 3-element front wings – if drag was measured on the front wing alone, a 3-element front wing would undoubtedly produce greater induced and pressure drag (a 3-element front wing produces greater downforce); however, measuring the total drag on both front wing and wheels would account for the increased diversion of airflow over the wheels (and hence reduced drag) caused by a greater number of flaps, providing a more realistic comparison. This will be discussed further in Section 3.3.

(a) (b)

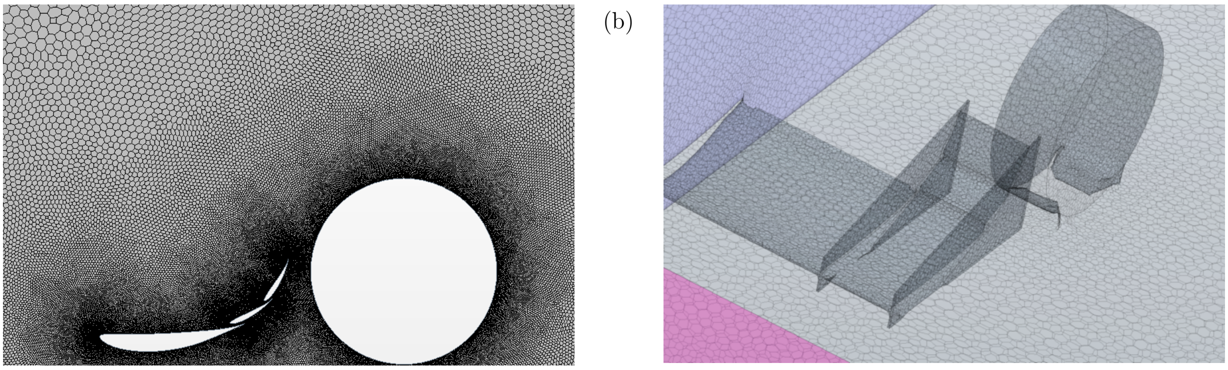

Figure 4: (a) Mesh generated for 2D simulations on the front wing and wheel. (b) Mesh generated for 3D simulations on the front wing and wheel – a symmetry plane allowed only half the front wing to be simulated. The wheel was made to intersect the ground to avoid problems with meshing the interface between the wheel and the ground.

Following the simulations run on the front wing, full-car 3D CFD simulations were carried out to visualise the interactions between different aerodynamic components and to check for aerodynamic balance. Figure 5 shows the setup of the CFD domain around the full car. Appropriate distances were specified between the car and the domain boundaries to allow for wake development behind the car and subsonic propagation of disturbances ahead of and around the car. A plane of symmetry allowed half the car to be simulated, approximately halving the computational cost of each simulation. For increased accuracy, the ground velocity was specified at 16.5 m/s relative to the car and the wheels were prescribed a rotation rate of 65 rad/s.

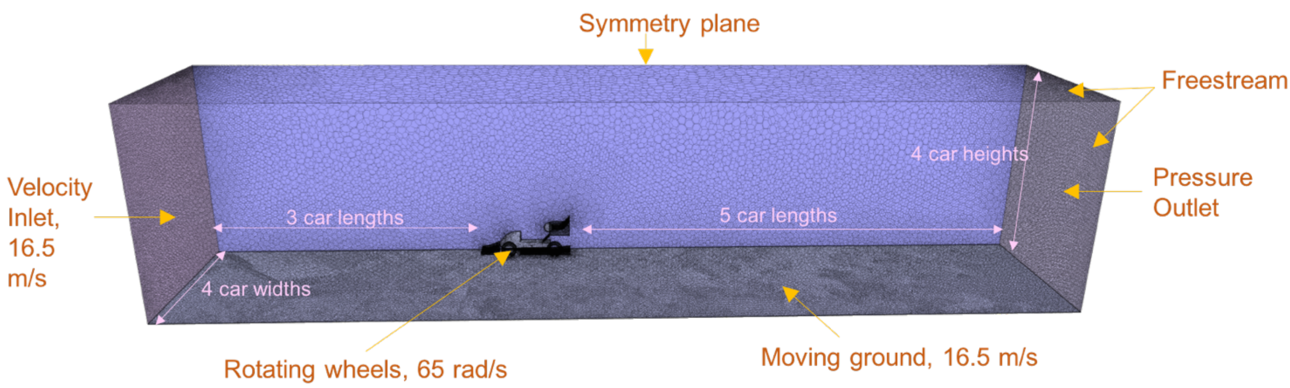

Figure 5: Setup of the CFD domain around the full car, showing dimensions and boundary conditions.

The following physics models were specified:

- K-ω SST Turbulence model: This model was chosen due to its ability to effectively predict flows under adverse pressure gradients; this is because it combines the best of two models – it solves the k-ω equations inside boundary layers and the standard k-ϵ equations futher away from walls [8]. To substantiate this argument, the k-ω SST model was compared to the k-ϵ model during CFD validation (see Figure 6), where it performed better.
- Steady flow: The unsteady effects of shed vortices were neglected and a steady flow was specified in order to reduce simulation times.
- Segregated flow: The segregated flow solver was chosen over the coupled flow solver as the coupled solver utilises more memory [9] and requires greater simulation times.

- Turbulence intensity: The turbulence intensity of the flow at inlet was specified at 0.2% assuming the air in front of the car to be stationary. The turbulence intensity of the flow at outlet was however increased to 2% [10].

3.2.2 Validation

To ensure the validity of the CFD simulations, two solvers – the k-ω SST and the k-ϵ – were first used to calculate the C_L and C_D experienced by the Eppler E423 aerofoil in 2D (at Re=300000), as a function of angle of attack. The results were then compared to known wind tunnel data [11] as well as simulations run on XFoil, as shown in Figure 6.

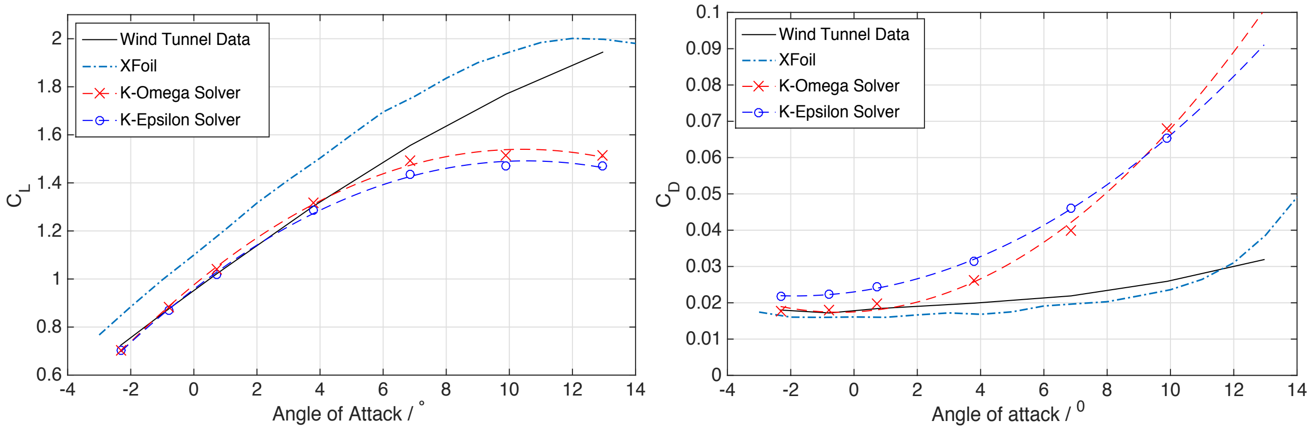

Figure 6: Variation of C_L and C_D with angle of attack, obtained from 2D CFD simulations employing both the k-ω SST and the k-ϵ solvers. These are compared to known wind tunnel data as well as results from XFoil.

The C_L values obtained using both CFD solvers showed good agreement to the tabulated wind tunnel data at low angles of attack ($-2°$ to $4°$), but diverged away at higher angles of attack. The XFoil data show an approximately 10% over-prediction in C_L when compared to the wind tunnel data, across all angles of attack. One possible explanation for this phenomena would be that at Re=300000, the physical aerofoil was under transition; XFoil may not have captured this leading to an overprediction in lift.

On the other hand, the C_D values from the wind tunnel and XFoil data matched well. However, both CFD solvers quickly over-predicted drag above an incidence of 2°, which is expected in 2D CFD simulations [12]; results are expected to improve in 3D. Finally, the k-ω SST solver was selected as it showed better accuracy when compared to the k-ϵ solver.

3.2.3 Mesh convergence

Following validation, a mesh convergence study was conducted in 2D to determine the minimum number of cells in the mesh that would give converged results. Tests were conducted on the Eppler E423 aerofoil at an incidence of 5° and the number of cells were varied by varying Base Size. The results are shown in Figure 7.

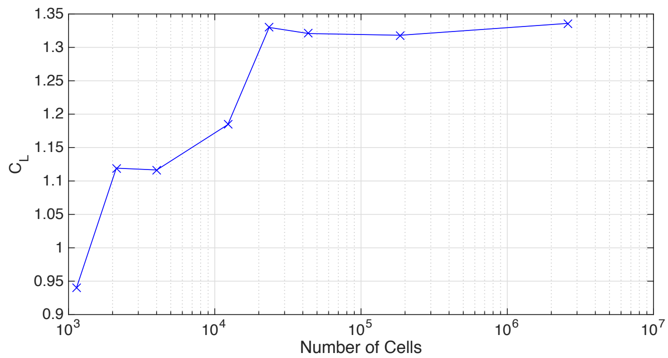

Figure 7: Measurements of C_L from 2D simulations run using different Base Sizes.

The measured value of C_L converged above 2×10^4 cells (corresponding to a Base Size of 0.05 m); this mesh density was thus chosen for all subsequent CFD runs.

3.3 Design using 2D Simulations

Wherever possible, 2D CFD simulations have been used to optimise design in order to save on computational time. The simulations were run on a cross-section of the front wing and front wheel. Measurements of downforce coefficient, C_L, were made on the front wing alone while measurements of drag coefficient, C_D, included both the front wing and the front wheel. All design decisions aimed to maximise C_L/C_D, or L/D.

3.3.1 Angle of attack & No. of elements

To decide between using a single, double or triple-element front wing, simulations were run on 2D models of the front wing in each of the three configurations with dimensions that spanned the entire legality box (in order to maximise chord lengths and hence downforce). In the single-element case, a single run was performed at the optimal mainplane angle of 5.25°, while in the double-element case, the mainplane angle was fixed and the flap angle varied to obtain an optimum, as shown in Figure 8. In the triple-element case, the mainplane angle was fixed while a sweep of both flaps' angles was carried out to produce the contour plot of L/D shown in Figure 9. The result of the study is summarised in Table 1.

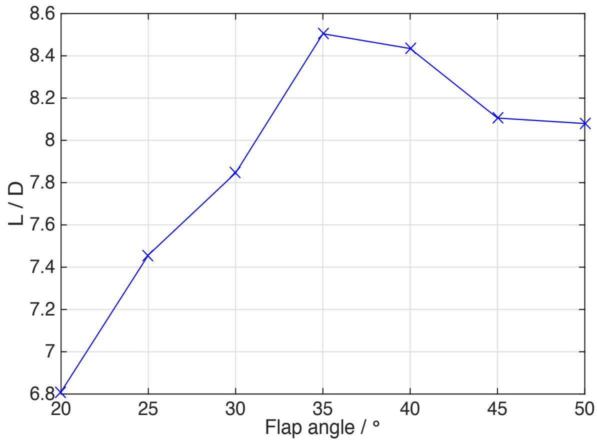

Figure 8: L/D variation with flap angle for a two-element front wing.

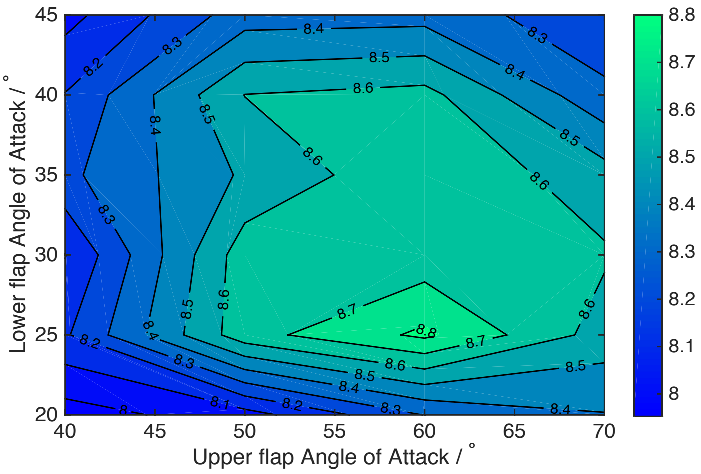

Figure 9: Variation of L/D (the contour lines) with both flap angles for a three-element front wing.

Table 1: Comparison between a single, double or triple-element front wing. Optimal angles of attack of each element are shown as well as the corresponding values of C_L, C_D and L/D.

No. of elements	Optimal mainplane angle / °	Optimal lower-flap angle/ °	Optimal upper-flap angle/ °	C_L	C_D	L/D
1	5.25	-	-	0.99	0.13	7.7
2	5.25	35	-	1.55	0.18	8.5
3	5.25	25	60	1.68	0.19	8.8

A triple-element front wing was chosen as it performed the best in terms of L/D and produced the highest downforce, which would aid aerodynamic balance with the rear of the car. From Figure 9, the optimal angles of attack of the mainplane and two flaps were chosen as 5.25°, 25° and 60° respectively.

The maximum downwards pitch of the car under braking was calculated to be 0.98°, knowing the bump (i.e. maximum downwards displacement of the car in the plane across the axle of the front wheel) for which the front-wheel suspension was designed [13]. Since the optimal angles chosen for the mainplane and flaps were not close to stall (an increase in incidence of 5° for each element did not lead to a decrease in lift), it was decided that there was no need for the setting angles to be reduced by 0.98° to avoid stall.

3.3.2 Ratio of chords

Following the determination of the setting angles of the 3 elements, the ratio of chord lengths taken by each element was decided. For simplicity, the chord lengths of the two flaps were taken to be equal. In order to maximise the space available, every combination of chord lengths tested spanned the entire regulation box, after taking ground clearance into account. Figure 10 shows the variation of L/D with mainplane chord length as a percentage of the total length.

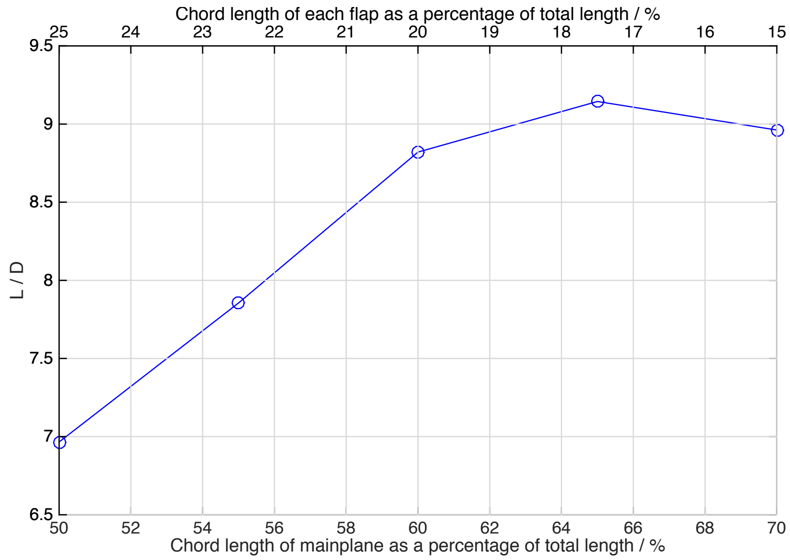

Figure 10: Variation of L/D with mainplane chord lengths. Also shown are the corresponding flap chord lengths.

A chord length ratio of 65% : 17.5% (corresponding to chord lengths of 336.9 mm : 90.4 mm) was thus chosen for the mainplane and flaps. The analysis can be improved in future iterations by varying the chord lengths of all three elements simultaneously to obtain a more accurate optimum.

3.3.3 Slot gaps

Slot gaps, i.e. the vertical distance between elements, allow a portion of the airflow over the high pressure side of the leading element to flow over the suction side of the succeeding element [14], effectively re-energising the boundary layer there and allowing it to stay attached. This allows for greater incidences and hence a greater downforce [15]. To achieve a convergent slot gap, a slight overlap between elements is needed, up to the thickest part of the succeeding element.

In order to save on computational time, the slot gaps between all elements were taken to be equal, although slot gaps between flaps are typically smaller than those between the mainplane and flap [16]. Figure 11 shows the result of varying slot gap on L/D when run in CFD – a slot gap of 12.5 mm was thus chosen.

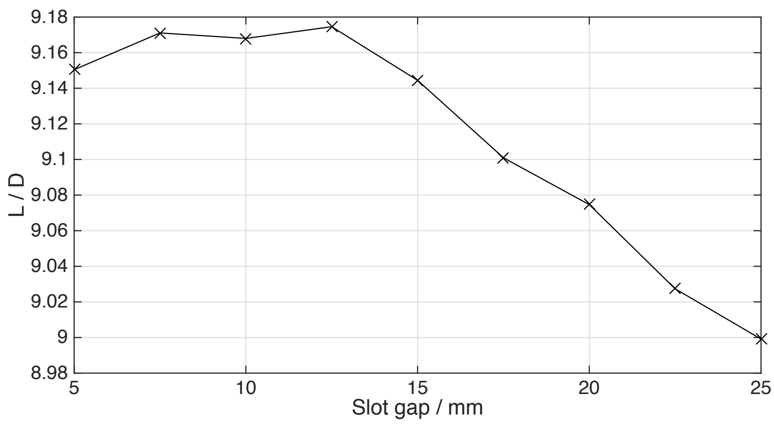

Figure 11: Variation of L/D with slot gap distance.

3.3.4 Ground clearance

The proximity of the front wing to the ground allows ground effect to be taken advantage of; the gap between the mainplane and the ground forms a Venturi-like cross-section, accelerating flow beneath the front wing and reducing static pressure – this increases downforce significantly. Generally, a smaller ground clearance translates to greater downforce. However, below a ground clearance of about 10% chord, the strong adverse pressure gradients developing in the channel between the front wing and the ground will lead to stall [17].

The maximum downwards travel of the front wing at its lowest point was calculated to be 36.7 mm using the maximum bump from the suspension team [13]. Hence a ground clearance of 40 mm was decided upon in order to avoid contact between the front wing and the ground – since this was close to being 10% of the mainplane chord, a CFD simulation was run with a ground clearance of 35 mm to check that the downforce had not decreased; that is, to check that the front wing would not stall at the prescribed ground clearance.

3.4 Design using 3D Simulations

Having decided on the cross-sectional parameters of the front wing, the three-dimensional characteristics were decided by running CFD simulations in 3D, with the model set up as shown in Figure 4(b). Measurements of downforce were again measured on the front wing alone while drag was measured on both the front wing and wheels. The front wing span was chosen to be 1445 mm to fully utilise the space within the regulation box.

3.4.1 End plates

Endplates are a common feature found on the front wings of most race cars due to the performance increase associated with their use. Endplates increase the effective aspect ratio of the relatively small aspect-ratio front wing and reduce its 3D effects by inhibiting flow leakage from the high-pressure surface to the suction surface [18]. A greater endplate area generally translates to improved performance, but also means greater weight. Two different endplate shapes were thus compared using CFD analysis: a rectangular one and a tapered one, as shown in Figure 12. The flap span (length of the flapped section of the front wing) was taken to be 200 mm for this analysis; a comparison of flap spans will be discussed in the next section.

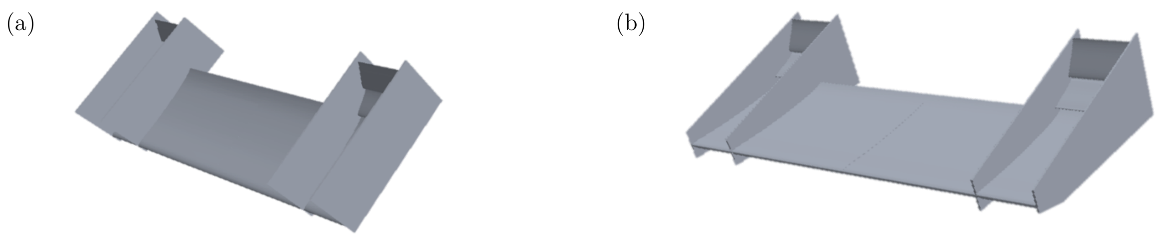

Figure 12: (a) Front wing with rectangular endplates. (b) Front wing with tapered endplates.

As can be seen in the figure above, there are four endplates: two on the outside and two inside. The dimensions of both endplates were chosen to just envelop the three elements of the front wing in the interest of conserving mass. The endplates were only allowed to extend 2 mm below the lowest point of the front wing to satisfy ground clearance – the lack of endplate area between the suction surface and freestream will inevitably lead to a decrease in lift and a higher induced drag [19]; future iterations of the front wing should consider finding an optimum between ground clearance and the extension of the endplate below the front wing. Table 2 outlines the results obtained from the CFD simulations.

Table 2: Comparison between rectangular and tapered endplates.

Endplate	Downforce / N	Drag / N	L/D
Rectangular	175.1	53.1	3.3
Tapered	171.5	52.0	3.3

Although using rectangular endplates led to a slightly greater downforce, the difference in L/D was negligible. To conserve weight, tapered endplates were chosen with a thickness of 6 mm to satisfy Regulation T9.5.

3.4.2 Gurney flaps

Gurney flaps, which are short strips fitted on the trailing edge of the high-pressure side of an aerofoil, are commonly used for their contribution to increasing downforce, albeit at a drag penalty. Measurements conducted using laser Doppler anemometry [20] revealed the mechanism behind how Gurney flaps increase downforce – they induce alternatively shed vortices causing a reduction in pressure on the suction side of an aerofoil. Furthermore, flow is decelerated over the high-pressure side, leading to an increase in pressure there.

A Gurney flap with a 10 mm × 10 mm cross-section was added onto the uppermost flap as shown in Figure 13 (regulations require a minimum thickness of 10 mm) and CFD tests were carried out to investigate its benefits, the results for which are detailed in Table 3.

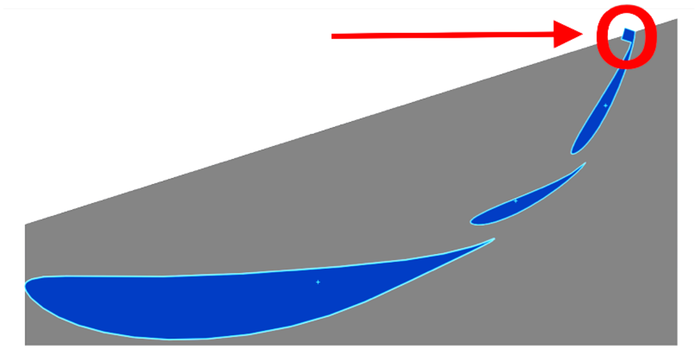

Figure 13: Cross-section of the front wing showing the addition of a 10 mm × 10 mm Gurney flap.

Table 3: CFD results for the front wing with and without a Gurney flap.

Configuration	Downforce / N	Drag / N	L/D
Without Gurney	171.5	52.0	3.3
With Gurney	173.9	53.9	3.2

Although using a Gurney flap produces a slight increase in downforce, the corresponding increase in drag caused an overall decrease in L/D; the Gurney flap was hence rejected.

3.4.3 Flap span

In order to study the variation in downforce and L/D resulting from different spans of the flapped section of the front wing, two flap spans were compared in CFD: 200 mm (which would cover just the front wheels) and 460 mm, as shown in Figure 14.

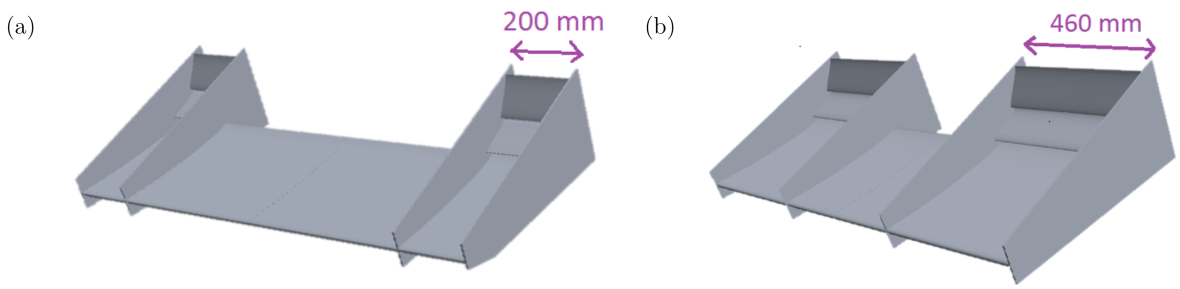

Figure 14: (a) Front wing with a flap span of 200 mm. (b) Front wing with a flap span of 460 mm.

Table 4 compares the downforce, drag and L/D values measured in the two tests.

Table 4: Comparison between 200 mm and 460 mm flap spans.

Flap span	Downforce / N	Drag / N	L/D
200 mm	171.5	52.0	3.3
460 mm	246.9	78.8	3.1

It can be seen that increasing the flap span from 200 mm to 460 mm increases the downforce by nearly 50% with a minimal decrease in L/D. For maximum downforce (which is needed for aerodynamic balance with the rear), the flap span was chosen to be 380 mm – this allowed a 65 mm gap between the inner endplates and the bodywork in order to channel airflow through to improve diffuser performance and aid cooling.

4 Full Car Analysis

Following the completion of the design of each aerodynamic component, a CFD simulation was run on the full car to visualise key interactions between components and measure more realistic downforce and drag values when accounting for these interactions. The model used in the simulation included only the major components affecting aerodynamics; smaller components of the car were neglected in order to allow a slightly coarser mesh and a shorter convergence time. Additionally, a CFD simulation was carried out on the car with its aerodynamic package removed in order to obtain the corresponding SC_L and SC_D values.

4.1 Interactions Between Components

As principal of the aerodynamics team, one of my main tasks was making sure that the development of all aerodynamic components were running in parallel. The latest estimates of downforce and drag were constantly obtained for each component to prevent the overdesigning of any single component as this would shift the aerodynamic balance far away from the centerline of the wheelbase. Satisfactory interactions between components were ensured, not just in terms of flow fields but geometrical constraints as well. For example, when the positions of the four wheels were shifted aft (to shift the center of gravity closer to the middle of the wheelbase), the front wing's mainplane chord length was reduced to 240 mm in front of the bodywork; this was necessary to maintain close proximity between the front wing and wheels while still accommodating the bodywork. Another example would be the truncation of the underside of the bodywork where the undertray-diffuser component began – this prevented an overlap and led to a reduction in mass.

Flow visualisation, despite not providing quantitative data, is an important tool in predicting performance and identifying critical areas for improvement. Figures 15(a) to (d) show some of the important interactions that were identified.

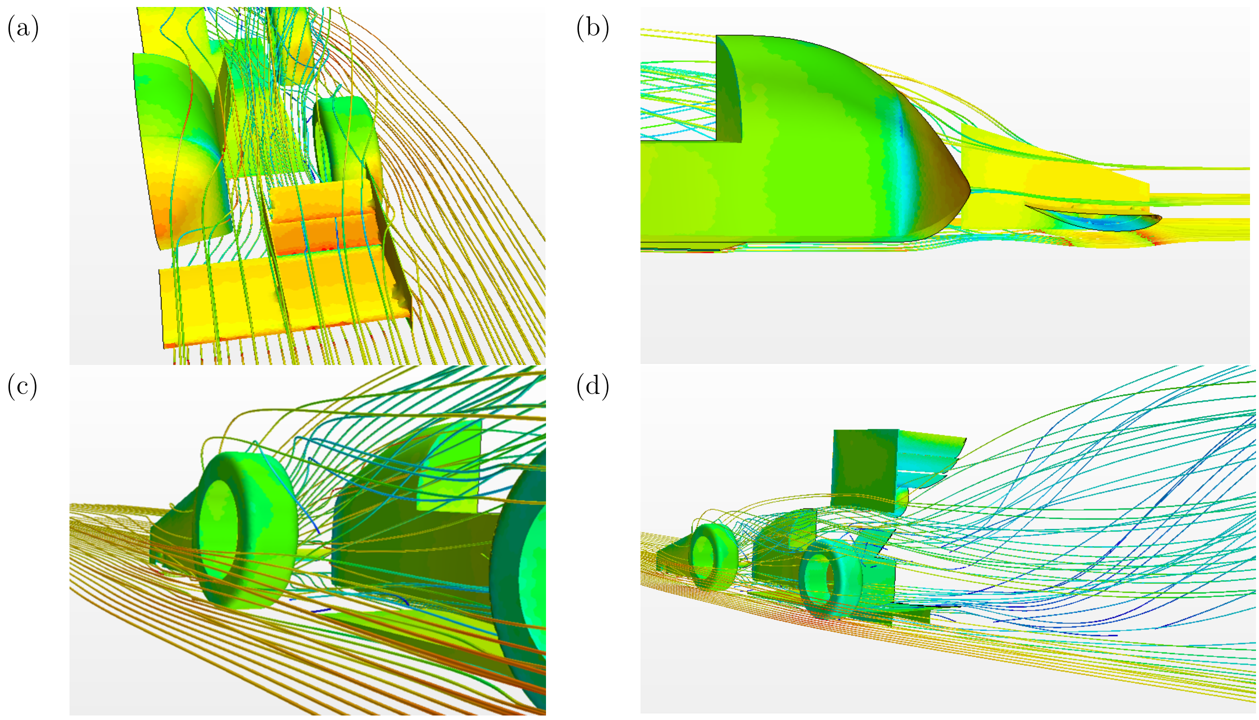

Figure 15: (a) to (d): Streamlines of airflow around the major aerodynamic components of the car. The contour on the car surface represents static pressure while that on the streamlines represent velocity, with red indicating a large magnitude and blue indicating a small one.

Figure 15(a) shows airflow being channeled through the 65 mm gap between the inner endplate and the bodywork. To ensure that the flow incident on the bodywork is laminar (for low drag), a single element mainplane was decided for the section of the front wing directly in front of the bodywork, as shown in Figure 15(b). The successful diversion of airflow over the front wheels by the front wing can be seen in Figure 15(c). Finally, Figure 15(d) shows the flow downstream of the diffuser being diverted upwards due to the low pressure region created behind the rear wing; the impact of this interaction was neglected due to the great distance between the rear wing and diffuser.

Following the visualisation of streamlines over the car, the important interactions between aerodynamic components were identified and a simplified N^2 diagram was created to illustrate them, as shown in Figure 16. In reality, however, almost all components will have an impact on one another.

Front Wing	x	x		x	x
o	Bodywork	o			x
	o	Diffuser			
			Rear Wing		x
xo		o	o	Wheels	x
					Cooling

Figure 16: N^2 diagram showing only the major aerodynamic ('x') and geometric ('o') interactions between components. The N^2 diagram is read in a clockwise direction, with a marker being placed between two interacting components; for example, the 'x' placed in cell [1,5] indicates that the flow over the front wing affects the wheels.

4.2 Mass Flow Rate to Radiators

Once the longitudinal position of the radiators have been determined, a plane showing velocity magnitudes was taken across that position in order to decide whether to place the radiators directly behind the driver's seat or to place it outboard, facing the incoming flow. It was decided that in order to receive the required mass flow rate, an outboard placement was necessary.

4.3 Results & Score Prediction

Table 5 shows the downforce and drag values measured in the full car simulation, while Table 6 shows the comparison of SC_L and SC_D values for the car with and without an aerodynamic package.

Table 5: Downforce and drag values measured from the full car CFD simulation. Note that 'Others' include all components of the car that are not part of the aerodynamic package. The downforce on the bodywork was measured to be negative because the measurements did not include its underbody.

Component	Downforce / N	Drag / N
Front Wing	186	36
Diffuser	152	6
Bodywork	-22	31
Rear Wing	219	67
Others	22	74
Total (full car)	**557**	**214**

Table 6: Comparison of dimensional coefficients for the car with and without an aerodynamic package.

	SC_L	SC_D
With Aero Package	3.34	1.29
Without Aero Package	0.07	0.74

The aerodynamic package produced an increase in SC_L of 3.27 at an SC_D penalty of +0.55. Running the lap simulator with the results of Table 6 and the location of the overall center of pressure of the car, a point gain of 36.4 was achieved with the inclusion of the aerodynamic package as shown in Table 7; this analysis took the weight of the aerodynamic package (25.3 kg) into account.

Table 7: Score comparison across all dynamic events between a full car with an aerodynamic package and one without.

Event	With Aero Package / points	Without Aero Package / points
Acceleration	43.8	36.2
Skid Pad	50	50
Sprint	124	137.4
Endurance	189.2	207.7
Efficiency	64	76.1
Total	**471** (out of 675)	**507.4** (out of 675)

4.4 Aerodynamic Balance

The position of the Center of Pressure (C.o.P.), which is the point through which the resultant forces act on the car, determines the aerodynamic balance and plays an important role in the handling characteristics of the car. Unevenly distributed aerodynamic loads will cause under-steer or over-steer when cornering [21], which is not desired. To prevent over-steer upon corner entry, a slight rear-bias of the Center of Pressure is required, especially for low-speed turns [22]. Figure 17 shows an estimate of the Center of Pressure position calculated using a moment balance equation that included the main downforce-generating components: the front wing, rear wing and diffuser. The load distribution on the wheel axles were found to be 60% : 40% rear-biased, which is beneficial to the car's rear-wheel drive configuration as it provides greater rear-wheel traction.

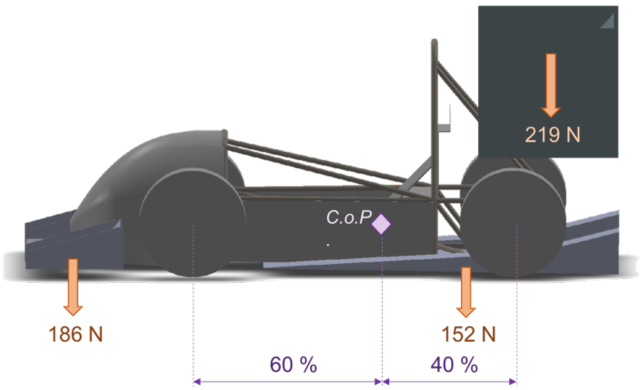

Figure 17: Center of Pressure position in relation to the wheelbase. Also shown are the downforce values on the front wing, rear wing and diffuser.

5 Manufacturing

5.1 Material Selection

The Cambridge Engineering Selector (CES) software was used in the selection of a suitable material for the manufacture of the front wing. To satisfy the regulations on minimum deflection, a plate assumption was used for the front wing; materials were ranked by ($E^{1/3}$/density) as shown in Figure 18, where E = Young's modulus. Cost was used in the horizontal axis because the overall budget was an important factor to consider.

Although natural wood gave the best performance in terms of ($E^{1/3}$/density), its use was decided against due to difficulties in making a shell out of a wooden block. The next highest performing material was carbon fibre, which was unfortunately too expensive. Thus an E-grade glass fibre composite with a layup of $[\pm 45°/0_2°/90°]_s$ was chosen for its relatively good performance and low cost; the outermost layers were chosen to be $\pm 45°$ in order to protect the 0° (spanwise) plies from damage.

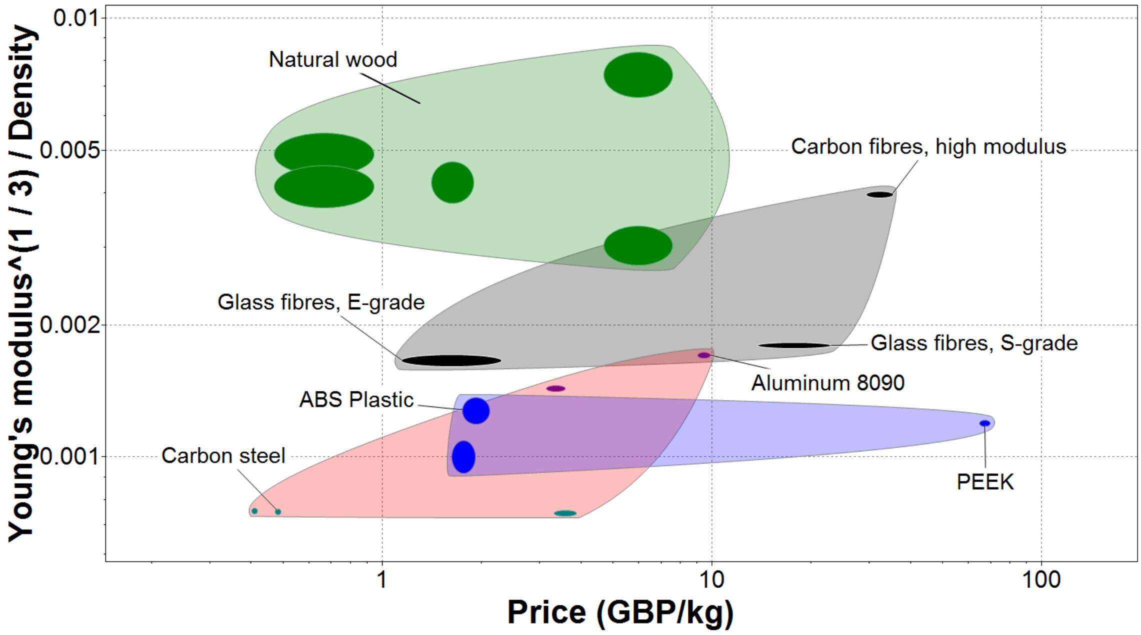

Figure 18: Ashby plot showing the criteria used in material selection.

The manufacturing method chosen for the front wing was as follows: A polystyrene mandrel carved in the desired shape is produced and a wax release agent applied onto it. A low-temperature epoxy resin is applied alternately with sheets of glass fibre while rollers are used to remove air bubbles and ensure material conformation. Depending on the curing temperature of the resin, the use of an oven may be required. The total weight of the front wing was calculated to be 7.50 kg and the cost of front wing manufacture was estimated to be GBP 396 [23].

5.2 Structural Analysis

The front wing will be attached to the chassis via two pylons 450 mm apart, bolted onto the front wing at the mid-chord of the mainplane [24], as shown in Figure A1 of Appendix A-1. A structural member will be added in between the two pylons to minimise sideways deflections of the front wing under loading conditions.

Regulation T9.7 requires that a load of 200 N applied anywhere on the front wing causes a deflection of no more than 25 mm. A Finite Element analysis was thus carried out on Solidworks to check that the regulation is met; the front wing of 2 mm thickness was clamped at the pylon attachment point and a pressure load (resulting to 200 N) was applied on the outer endplate of the front wing, furthest away from the attachment point. The maximum deflection measured was 17.2 mm, as shown in Figure A2 of Appendix A-1.

6 Conclusion

The inclusion of the full aerodynamic package for the EV4 showed an increase of 36.4 points in the dynamic events at a cost of GBP 2162. A favourable rear-biased aerodynamic balance of 60% : 40% was also achieved, which means that the front wing was successfully designed to match the downforce generated by the rear aerodynamic components of the car. A summary of the final design specifications of the front wing can be found in Table A1 of Appendix A-2 and a technical drawing in Appendix A-3. Future iterations of the front wing may include a Drag Reduction System, which involves reducing the incidence of the flaps by motor actuation; if this is used, the drag experienced on straights will decrease significantly and the front wing design can be aimed at maximising downforce instead of L/D. Given more time, the CFD analyses run in 2D can be replaced with 3D simulations to take 3D effects into account. Ultimately, the decision to invest the aforementioned amount of money into the implementation of an aerodynamic package will belong to the Imperial Racing Green team.

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Appendix

A-1 Structural Analysis

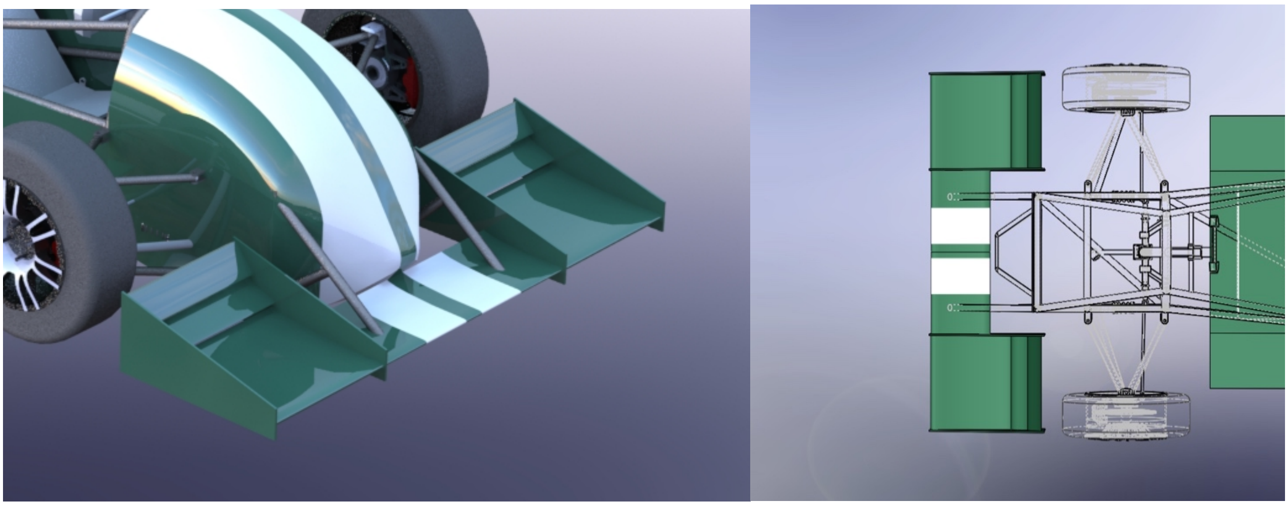

Figure A1: Position of pylon attachment to the front wing. Note that the details of the bolting mechanism between the pylon and the front wing are not shown.

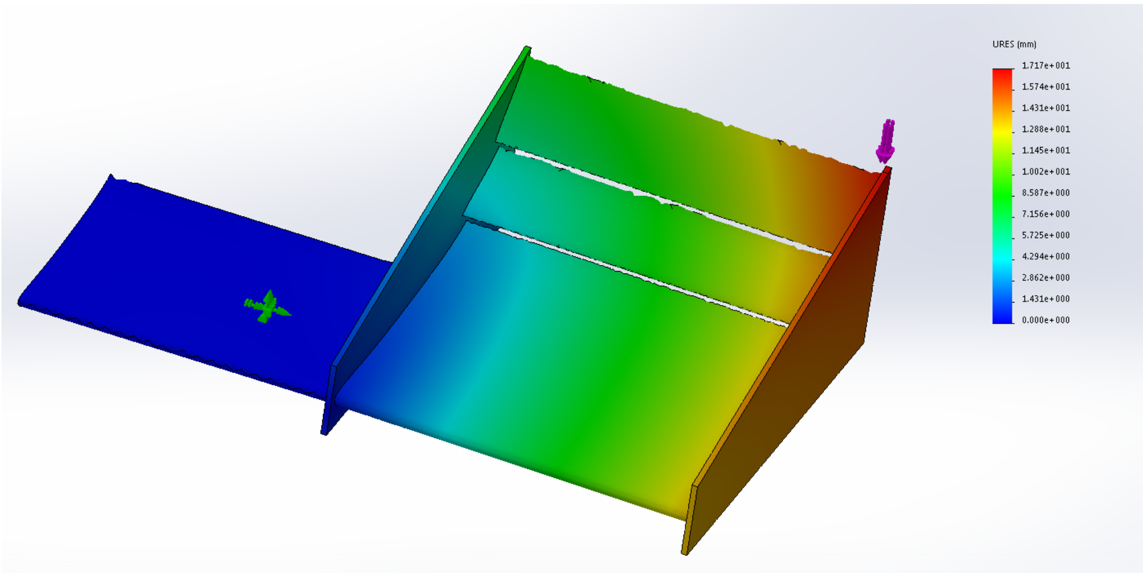

Figure A2: Results from the Finite Element analysis performed on the front wing. The contours show vertical displacement.

A-2 Final Design Specifications

Table A1: Summary of the final design specifications of the front wing.

Design parameter	Value
Mainplane aerofoil	FX76-MP-120
Flap aerofoil	Eppler E216
Number of front wing elements	3
Mainplane angle of attack	5.25°
Lower flap angle of attack	25°
Upper flap angle of attack	60°
Mainplane chord length (outside bodywork)	336.9 mm
Mainplane chord length (in front of bodywork)	240 mm
Chord length of each flap	90.4 mm
Slot gap	12.5 mm
Ground clearance	40 mm
Front wing span	1445 mm
Endplate	Tapered
Endplate thickness	6 mm
Gurney flaps	None
Flap span	380 mm

A-3 Technical Drawing

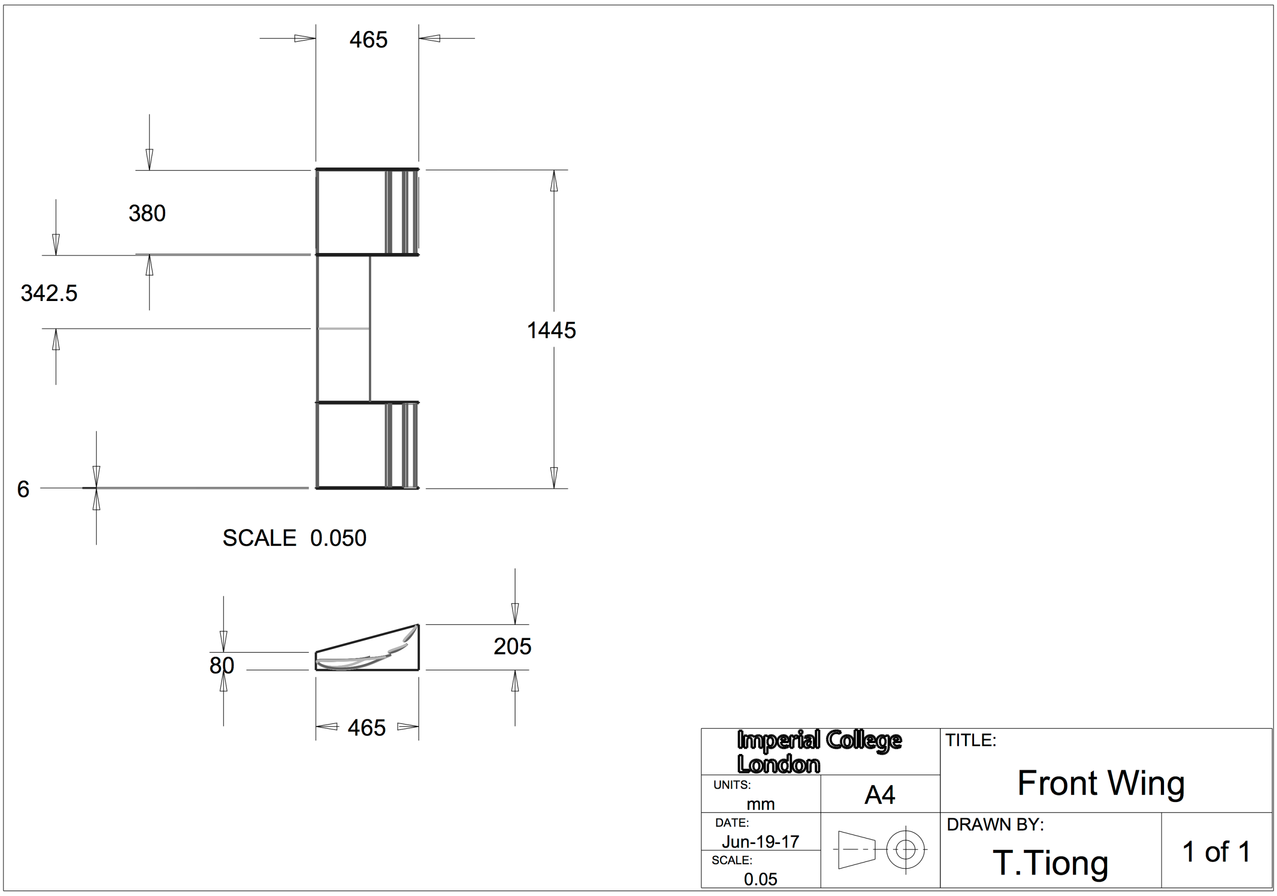

Figure A3: Technical drawing of the front wing.